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Category

Lesson 5: Probability

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Dependent Events

Two events are dependent when the first event affects the second event.

Example

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There are 10 marbles in a bag: 7 are red and 3 are blue. Lisa draws a marble out of the bag. Without replacing the first marble, she draws a second marble from the bag. What is the chance that both marbles are blue?

Step 1

Find the probability of the first event.

$$\frac{3 \text{ blue}}{10 \text{ marbles}} = \frac{3}{10}$$

Step 2

Assume Lisa has drawn out a blue marble. Now there are 9 marbles left and 2 are blue. Find the probability of the second event.

$$\frac{2 \text{ blue}}{9 \text{ marbles}} = \frac{2}{9}$$

Step 3

Multiply the probabilities.

$$\frac{3}{10} \times \frac{2}{9} = \frac{6}{90} = \frac{1}{15} \approx 7\%$$

The probability of drawing 2 blue marbles is **1 out of 15**, or **about 7%**.

If Lisa replaces the marble before drawing out the second marble, the events would be independent. Do you see why?

The number of blue and total marbles would not change. If she replaces the marble, the probability is $\frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$,

or 9%.

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